

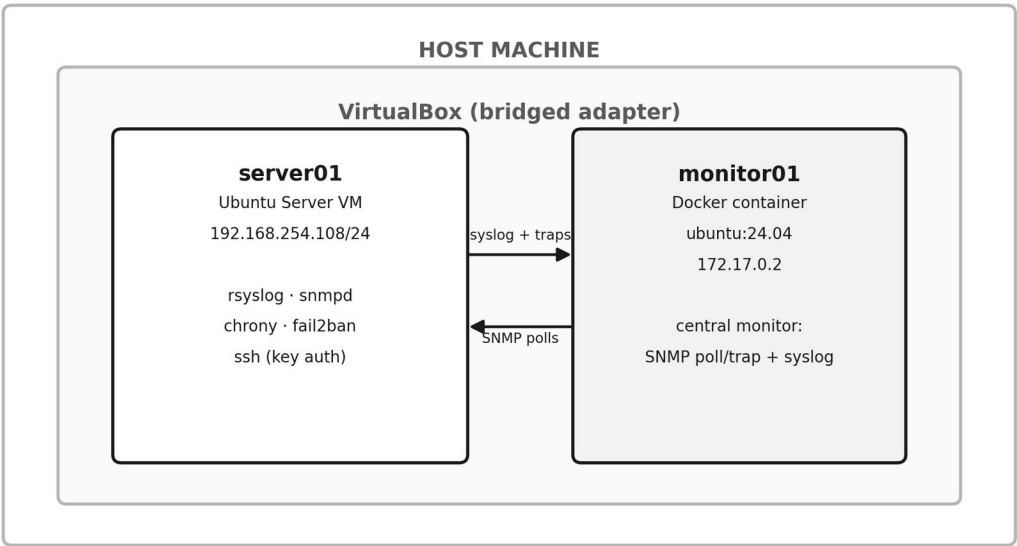
VirtualBox + Docker Home Lab

By Dylan Andrei P. Kiocho

Objective

A two-host home lab built with VirtualBox and Docker: server01 (Ubuntu Server) running core services, and monitor01 (a Docker container) acting as a central monitoring and log-collection point for it. The lab covers static IP setup, SNMP-based monitoring, centralized logging, time sync, and login protection — built step by step, with most steps hitting a real problem that had to be diagnosed.

Architecture



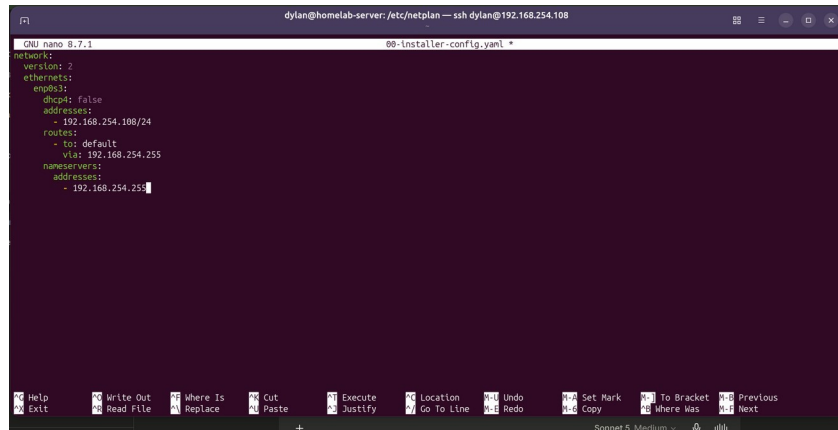
server01 runs the core services (rsyslog, snmpd, chrony, fail2ban, SSH). monitor01 is a Docker container that polls server01 over SNMP and receives its logs and alerts — one host does the work, the other watches it.

Environment

Component	Details
Hypervisor	VirtualBox, bridged adapter
server01	Ubuntu Server, static IP
monitor01	Docker container (ubuntu:24.04)
Stack	net-snmp, rsyslog, chrony, fail2ban

Setup

server01 was moved from DHCP to a static IP so the rest of the lab would have a fixed target to work against. SSH access was locked down to key-based login only, with root restricted to keys as well. SNMP read access was configured for both v2c and v3, and monitor01 was stood up as a Docker container reachable from server01 in both directions.

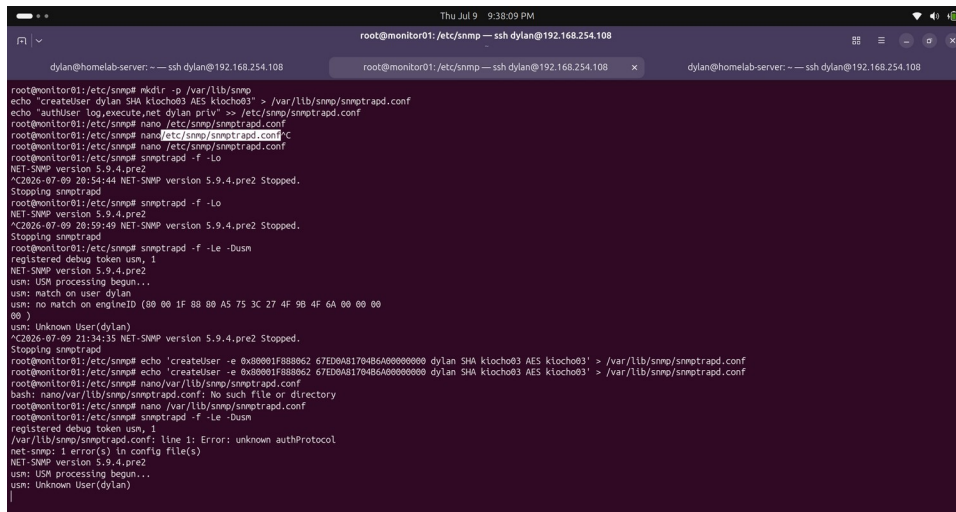


Static IP configured in netplan.

SNMP Monitoring — Poll and Trap

This was the core of the project. monitor01 was set up to both poll server01 for metrics (SNMP poll) and receive push alerts from it (SNMP trap), under SNMPv3 for authentication and encryption.

The trap side broke first: server01's messages kept getting rejected. Debug output showed the receiving side was rejecting the engineID — a unique identifier tied to server01 that has to match exactly for SNMPv3 to trust the message. A packet capture confirmed the traffic itself was arriving fine, which narrowed the problem down to that one mismatched value. Reading server01's real engineID directly fixed it.



Debug output showing the engineID rejection.

Polling had a separate issue — timeouts, then an access error, before the right target and permissions were sorted out and a clean poll went through. Getting both working end to end, and separately, made the difference between poll and trap.

```
^C2026-07-11 01:11:27 NET-SNMP version 5.9.4.pre2 Stopped.
Stopping snmptrapd
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 172.17.0.2 .1.3.6.1.2.1.1.3.0
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 172.17.0.2 .1.3.6.1.2.1.1.3.0
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 172.17.0.2 .1.3.6.1.2.1.1.3.0
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 172.17.0.2 .1.3.6.1.2.1.1.3.0
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1.3.0
snmpget: Timeout
root@monitor01:~# snmpwalk -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1
snmpwalk: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 127.0.0.1 .1.3.6.1.2.1.1
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 127.0.0.1 .1.3.6.1.2.1.1
snmpget: Timeout
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1
Error in packet
Reason: authorizationError (access denied to that object)
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 1.3.6.1.2.1.25.3.3.1.2
Error in packet
Reason: authorizationError (access denied to that object)
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 1.3.6.1.2.1.25.3.3.1.2
Error in packet
Reason: authorizationError (access denied to that object)
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 1.3.6.1.2.1.25.3.3.1.2
iso.3.6.1.2.1.25.3.3.1.2 = No Such Instance currently exists at this OID
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1
iso.3.6.1.2.1.1 = No Such Object available on this agent at this OID
root@monitor01:~# snmpget -v 3 -u dylan -l authPriv -a SHA -A kIocho03 -x AES -X kIocho03 192.168.254.108 .1.3.6.1.2.1.1.1.0
iso.3.6.1.2.1.1.1.0 = STRING: "Linux homelab-server 7.0.0-27-generic #27-Ubuntu SMP PREEMPT_DYNAMIC Thu Jun 18 19:13:49 UTC 2026 x86_64"
root@monitor01:~# clear
```

Poll troubleshooting: timeouts, then an access error, then a clean read.

```
root@monitor01:~# cd /var/lib/snmp
root@monitor01:~# snmptrapd -f -l
NET-SNMP version 5.9.4.pre2
2026-07-11 01:02:29 <UNKNOWN> [UDP: [172.17.0.1]:48945->[172.17.0.2]:162]:
iso.3.6.1.2.1.1.3.0 = Timeticks: (8192831) 22:45:28.31 iso.3.6.1.6.3.1.1.4.1.0 = OID: iso.3.6.1.4.1.2021.251.1
2026-07-11 01:05:47 <UNKNOWN> [UDP: [172.17.0.1]:48633->[172.17.0.2]:162]:
iso.3.6.1.2.1.1.3.0 = Timeticks: (8212544) 22:48:45.44 iso.3.6.1.6.3.1.1.4.1.0 = OID: iso.3.6.1.4.1.2021.251.1
```

A trap received cleanly after the fix.

Centralized Logging (rsyslog)

server01 was configured to forward its logs to monitor01, which automatically stores them in a separate folder per source host. A test message sent from server01 showed up on monitor01 correctly tagged with server01's hostname, confirming the forward was real and not just a local write. Both TCP and UDP forwarding were enabled, but TCP is the one that matters for anything worth not losing.

Break/Fix Testing

A few things were deliberately (and in one case accidentally) broken to practice real troubleshooting instead of just confirming a clean setup:

- SSH access dropped intermittently between the host and server01 — traced to a Wi-Fi networking quirk, still an open issue with a workaround in place.
- server01's clock drifted out of sync, which actually blocked a package update until it resynced — confirmed fixed via a clean time-sync check afterward.

```
dylan@homelab-server: /etc/netplan -- ssh dylan@192.168.254.108
dylan@homelab-server: /etc/netplan -- ssh dylan@192.168.254.108
dylan@homelab-server: ~ -- ssh dylan@192.168.254.108

nameservers:
  addresses:
    192.168.254.254
dylan@homelab-server: /etc/netplan$ sudo netplan apply
dylan@homelab-server: /etc/netplan$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:ff:fe:b3:3e brd ff:ff:ff:ff:ff:ff
    altname enx080027ffb33e
    inet 192.168.254.108/24 brd 192.168.254.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27fff:feb3:3e/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
default via 192.168.254.254 dev enp0s3 proto static
192.168.254.0/24 dev enp0s3 proto kernel scope link src 192.168.254.108
dylan@homelab-server: /etc/netplan$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:ff:fe:b3:3e brd ff:ff:ff:ff:ff:ff
    altname enx080027ffb33e
    inet 192.168.254.108/24 brd 192.168.254.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27fff:feb3:3e/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
dylan@homelab-server: /etc/netplan$ sudo apt update && upgrade
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Error: Release file for http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease is not valid yet (invalid for another 2d 15h 7min 4s). Updates for this repository will not be applied.
Error: Release file for http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease is not valid yet (invalid for another 2d 15h 6min 53s). Updates for this repository will not be applied.
```

The clock drift blocking a real package update.

```
Sat Jul 11 12:40:56 AM
dylan@homelab-server: ~ -- sudo ssh dylan@192.168.254.108
dylan@homelab-server: ~ -- sudo ssh dylan@192.168.254.108
root@monitor01: /var/log/homelab-server -- ssh dylan@192.168.254.108

dylan@homelab-server:~$ date
Fri Jul 10 04:40:26 PM UTC 2026
dylan@homelab-server:~$ sudo chronyc tracking
[sudo: authenticate] Password:
Reference ID      : 7023300 (112.210.51.208.pldt.net)
Stratum          : 2
Ref time (UTC)   : Fri Jul 10 16:40:46 2026
System time      : 0.00128061 seconds slow of NTP time
Last offset      : -0.000004091 seconds
RMS offset       : 0.002977934 seconds
Frequency        : 13979.147 ppm slow
Residual freq    : -0.217 ppm
Skew             : 3.866 ppm
Root delay       : 0.00135025 seconds
Root dispersion  : 0.000435652 seconds
Update interval  : 64.8 seconds
Leap status      : Normal
dylan@homelab-server:~$
```

Confirmed back in sync afterward.

- A real login lockout was triggered and resolved through fail2ban — ban, block, unban, and access restored.

```
homelab-server [Running] - Oracle VirtualBox
[ 10.124003] vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
[ 10.124005] vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.

homelab-server login: dylan
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Jul 9 07:40:31 AM UTC 2026

System load:  0.39           Processes:    127
Usage of /:   19.7% of 19.5GB Users logged in:   0
Memory usage: 20%           IPv4 address for enp0s3: 192.168.254.108
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

dylan@homelab-server:~$ [45102.912166] e1000 0000:00:03:00 enp0s3: Reset adapter
dylan@homelab-server:~$ sudo fail2ban-client status sshd
[sudo: authenticate] Password:
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed: 6
| -- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
|- Actions
| |- Currently banned: 1
| |- Total banned: 1
| -- Banned IP list: 192.168.254.113
dylan@homelab-server:~$
Display all 175 possibilities? (y or n)
dylan@homelab-server:~$
```

A real fail2ban lockout in action.

```
dylan@homelab-server: ~ — sudo ssh dylan@192.168.254.108
dylan@homelab-server: ~ — sudo ssh dylan@192.168.254.108
root@monitor01: /var/log/homelab-server — ssh dylan@192.168.254.108

dylan@dylan-Nitro-AM515-44:~$ ssh dylan@192.168.254.108
ssh: connect to host 192.168.254.108 port 22: Connection refused
dylan@dylan-Nitro-AM515-44:~$ ssh dylan@192.168.254.108
ssh: connect to host 192.168.254.108 port 22: Connection refused
dylan@dylan-Nitro-AM515-44:~$ sudo ssh dylan@192.168.254.108
[sudo: authenticate] Password:
ssh: connect to host 192.168.254.108 port 22: Connection refused
dylan@dylan-Nitro-AM515-44:~$ sudo ssh dylan@192.168.254.108
dylan@192.168.254.108's password:
dylan@192.168.254.108:~$
Welcome to Ubuntu 22.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Jul 10 04:37:32 PM UTC 2026

System load: 0.0          Processes: 120
Usage of /:  28.8% of 19.51GB    Users logged in: 0
Memory usage: 22%             IPv4 address for enp0s3: 192.168.254.108
Swap usage: 0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Jul 10 05:53:52 2026 from 192.168.254.113
dylan@homelab-server:~$
```

Access restored after unbanning — same IP that was blocked, now let back in.

Status

Core lab is complete: networking, monitoring, logging, and break/fix testing all working and verified.